HydroStock

New User Guide, Navigation Dictionary and Role Workflow

Purpose: Help a new user understand what each navigation option does, what every important field means, and how the roles work together.

Application URL: http://localhost:8080/HydroStock

System: Hydroelectric plant inventory, procurement, billing, supplier, and maintenance management.

Format: Times New Roman, reviewer-friendly project guide.

Read Sections 1–4 first for orientation. Use Sections 5–8 while demonstrating the application.

1. What HydroStock Does

HydroStock manages the complete flow from plant spare-part inventory to purchase, billing, supplier delivery, and payment. It also stores plant equipment and maintenance information.

Material Need → Stock / Low Stock → Purchase Order → Approval → Supplier Dispatch → GST Invoice → Payment → Reports

Business area	Simple explanation	Main records
Administration	Controls users, equipment, alerts, and system reports.	Users, roles, equipment, reports.
Inventory	Tracks what spare parts exist, where they are stored, and how stock changes.	Materials, categories, stock transactions.
Purchase and billing	Creates purchase orders, calculates GST, generates invoices, and records payments.	Suppliers, purchase orders, invoices, payments.
Supplier portal	Allows a supplier to see assigned orders, update dispatch status, and view invoices.	Supplier orders, dispatch updates, supplier invoices.

2. Login Roles and Their Relationship

Role	Who uses it	Navigation visible	Primary responsibility
ADMIN	Chief Engineer / system administrator	Dashboard, User Management, Plant Equipment, System Reports	Manage system users and plant-level administration.
INVENTORY_MANAGER	Store In-Charge	Material Catalog, Categories, Stock Inward, Stock Outward, Alerts, History	Maintain stock accuracy and issue/receive materials.
PLANT_ENGINEER	Maintenance or turbine engineer	Inventory and stock operation pages	Request/use material for plant maintenance and inspect stock history.
PURCHASE_MANAGER	Accounts or purchasing officer	Create PO, Purchase Orders, GST Bill, Invoices & Payments	Buy required materials, generate tax invoices, and record payment.
SUPPLIER	Registered vendor such as BHEL	Supplier Overview, Purchase Orders, Supplier Bills	Acknowledge orders and update dispatch/delivery information.

> **Important relationship:** Admin manages access. Inventory identifies the need and records stock. Purchase converts the need into a supplier order. Supplier updates delivery. Billing creates the statutory invoice. Accounts records payment. Reports combine the information.

Demo login accounts

Username	Password	Role
`admin`	`admin123`	ADMIN
`store_officer`	`store123`	INVENTORY_MANAGER
`purchase_officer`	`purchase123`	PURCHASE_MANAGER
`bhel_supplier`	`supplier123`	SUPPLIER
`turbine_tech`	`tech123`	PLANT_ENGINEER

3. Navigation Dictionary

Navigation option	What the user sees	When it is used
Dashboard	KPI cards, chart, notifications, low-stock list.	Quick overview of plant operations.
User Management	Users, role, status, email, phone, add/edit/delete actions.	Create or maintain application accounts.
Plant Equipment	Equipment list and maintenance log form.	Record equipment issues and maintenance work.
System Reports	Inventory, purchase, invoice, and transaction reports.	Review totals and prepare project or management reports.
Material Catalog	Material code, name, category, stock, price, rack, search.	Maintain the master list of plant spares.
Material Categories	Category name and description.	Group materials by turbine, valve, generator, oil, sensor, or fastener type.
Stock Inward	Receive quantity against a reference/challan.	Increase stock after material arrives.
Stock Outward	Issue quantity for equipment or purpose.	Decrease stock when material is used or issued.
Low Stock Alerts	Materials at or below minimum stock level.	Identify what should be purchased.
Stock History	Inward/outward transaction ledger.	Audit who changed stock, how much, and why.
Create Purchase Order	Supplier, delivery date, line items, quantity, price, notes.	Convert a material requirement into a formal order.
Purchase Orders	PO number, supplier, dates, value, status, actions.	Approve and monitor the procurement lifecycle.
Generate GST Bill	PO selection, Intra-State or Inter-State GST.	Generate the tax invoice after an order is ready.
Invoices & Payments	Invoice ledger and payment modal.	View invoices and record supplier payment.
Supplier Overview	Supplier KPIs and recent orders.	Supplier home page.
Supplier Purchase Orders	Assigned orders and dispatch modal.	Supplier reports DISPATCHED or DELIVERED status.
Supplier Bills	Supplier invoice list and tax invoice links.	Supplier reviews billing and payment records.

4. Field Dictionary: What to Enter

Material and category fields

Field	Meaning	Example
Material Code	Unique short identifier for a spare.	`MAT-TURB-001`
Material Name	Full technical description.	Francis Turbine Guide Bearing Pad
Category	Technical group used for filtering and reporting.	Turbine Spares & Bearings
Unit of Measure	How the item is counted.	Units, Sets, Pieces, Drums
Current Stock	Quantity physically available now.	`10`
Minimum Stock Level	Safety threshold that creates a low-stock alert.	`5`
Unit Price	Approximate or standard price per unit.	`85000.00`
Location Rack	Physical store location.	`Rack-A1-Bay02`

Stock operation fields

Field	Meaning	Required rule
Material	Item being received or issued.	Must be selected from live catalog.
Quantity	Number received or issued.	Positive integer, minimum 1.
Reference / Challan No.	Delivery challan, issue slip, work order, or reference.	Cannot be blank.
Remarks / Equipment Purpose	Why stock moved and where it was used.	Required for outward operations.

Purchase order fields

Field	Meaning	Example / rule
Registered Supplier	Vendor receiving the PO.	Select an existing supplier.
Expected Delivery	Date by which the material should arrive.	Today or a future date.
Material Line Item	Spare part being ordered.	Select from catalog.
Quantity	Number of units ordered.	Minimum 1.
Unit Price	Price per ordered unit.	Greater than 0.
Notes	Special delivery, certification, or quality instructions.	Optional but useful.

GST invoice and payment fields

Field	Meaning	Rule
GST Type	Intra-State or Inter-State supply.	Intra-State = CGST 9% + SGST 9%; Inter-State = IGST 18%.
Subtotal	PO value before tax.	Calculated from PO.
Grand Total	Subtotal plus GST.	Must equal Subtotal + Tax.
Amount Paid	Money released to supplier.	Positive and no more than pending balance.
Payment Mode	NEFT/RTGS, cheque, bank transfer, or treasury bill.	Select one valid mode.
Transaction / UTR Reference	Bank or payment trace number.	Required for audit.

5. Complete Demonstration Workflow

1. **Admin setup:** Log in as Admin, review users, roles, equipment, and reports.
2. **Inventory setup:** Log in as Store In-Charge, check the material catalog and low-stock alerts.
3. **Receive stock:** Open Stock Inward, select material, enter positive quantity, challan number, and remarks, then submit.
4. **Issue stock:** Open Stock Outward, select material, enter quantity and equipment/purpose. The system rejects a quantity greater than current balance.
5. **Create requirement:** From low-stock results, the purchase user opens Create Purchase Order.
6. **Raise PO:** Select supplier, set future delivery date, add material line items, quantity, unit price, and notes.
7. **Approve PO:** The purchase order status progresses through PENDING, APPROVED, DISPATCHED, and DELIVERED.
8. **Supplier update:** Supplier opens Purchase Orders and submits dispatch status with carrier/docket tracking notes.
9. **Generate invoice:** Purchase user selects the PO, chooses GST type, verifies tax preview, and generates the invoice.
10. **Record payment:** Accounts enters amount, mode, UTR/reference, and notes. The system updates UNPAID, PARTIALLY_PAID, or PAID.
11. **Review:** Admin opens reports and dashboard to review the resulting operational totals.

6. Status and Data Relationships

Record	Typical status values	Relationship
User	ACTIVE / INACTIVE	User receives one role and role-specific navigation.
Material	Low stock based on current stock and minimum stock	Belongs to one category and appears in stock, PO, and maintenance dropdowns.
Purchase Order	PENDING → APPROVED → DISPATCHED → DELIVERED	Belongs to one supplier and contains one or more material items.
Invoice	UNPAID / PARTIALLY_PAID / PAID	Generated from a PO and connected to payments.
Stock Transaction	INWARD / OUTWARD	Changes material balance and creates an audit history entry.
Maintenance Record	SCHEDULED / IN_PROGRESS / COMPLETED	Belongs to equipment and may reference material used.

7. Quick Troubleshooting Guide

Problem	Check first
Login page appears repeatedly	Confirm the user is active and the correct role credentials are being used.
Database-backed lists are empty	Start MySQL, import `database/database.sql`, and verify database name `hydrostock_db`.
Application is unavailable	Start Tomcat and run `build_and_deploy.bat`.
403 Forbidden	The account is not authorized for that module; log in with the role responsible for it.
Stock outward rejected	Reduce quantity to the current balance or below and provide a valid equipment/purpose remark.
PO cannot be submitted	Select a supplier, choose a future delivery date, add at least one item, and use quantity/price above zero.
Payment rejected	Check the pending balance and provide a positive amount plus UTR/reference number.

8. Project Review Summary

A new reviewer can understand HydroStock by following the role order: Admin setup, Inventory requirement, Purchase order, Supplier delivery, GST invoice, Payment, and Reports. Every role has only the navigation needed for its responsibility, while the database relationships connect the work across modules.

Application: `http://localhost:8080/HydroStock`

Companion document: HydroStock Testing and Implementation Report.